SWAP

KIOSK + WEB + MOBILE

FUNCTIONAL SPECIFICATION DOCUMENT

reachvel

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

Attribute	Details
Document Version	1.0
Prepared For	FitBar Technologies Pvt. Ltd | Swap
Document Type	Functional Specification Document (FSD)
Scope	Kiosk App, Website, Mobile Apps (iOS & Android)
Prepared Date	02 March 2026
Status	Draft — For Review
Developed By	Reachvel

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

1. Document Overview

This Functional Specification Document (FSD) defines the complete technical requirements, user flows, module breakdowns, data models, API integrations, and development action items for the FitBar platform. The platform comprises three components:

- Kiosk App (iPad-based, self-ordering terminal)
- Website & Meal Subscription Storefront
- Mobile Applications (iOS & Android)

> **Primary Integrations**
> Apex Commerce API (customer data & profiles) | Rista POS (inventory & order sync) | WhatsApp Business API (receipts & feedback)

1.1 Key Success Metrics

Metric	Target	Owner
Login to Checkout Time	< 30 seconds	Product / UX
Average Order Value (AOV)	Maximize	Product / Business
Customer Engagement Time at Kiosk	Minimize	UX
Upsell Conversion Rate	Track & Optimize	Analytics
System Uptime	> 99.5%	DevOps / Backend

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

2. System Architecture & Technology Stack

2.1 High-Level Architecture

The FitBar platform is a multi-client system backed by a shared REST/GraphQL API layer. All three frontends (Kiosk, Web, Mobile) communicate with a central backend service that interfaces with Rista POS and Apex Commerce.

Layer	Technology Recommendation	Notes
Kiosk Frontend	React Native (iPad) or Swift UIKit	Self-restart, offline-resilient
Web Frontend	Next.js (React)	SSR for SEO on landing pages
Mobile Apps	React Native (iOS + Android)	Shared codebase
Backend API	Node.js (Express/Fastify) or Django REST	Centralized service layer
Database	PostgreSQL (primary) + Redis (cache/sessions)	
POS Integration	Rista POS API (REST)	Real-time inventory sync
Commerce Integration	Apex Commerce API	Customer profiles, preferences
Payments	UPI via Razorpay / PhonePe Business	QR code-based
Messaging	WhatsApp Business Cloud API (Meta)	Receipts, OTP, feedback
Admin Panel	React + REST	Meal mgmt, analytics dashboard
Hosting	AWS / GCP (containerized with Docker + K8s)	

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

3. Module A — Kiosk Application

Platform: iPad | Framework: React Native or Swift | Environment: In-store, unattended

3.1 User Authentication & Profile Management

3.1.1 Functional Requirements

- Mobile Number + OTP login only. No guest checkout.
- OTP delivered via WhatsApp Business API (primary) with SMS fallback.
- On first login, prompt user to complete profile (name, height, weight, dietary goal).
- Profile saved to Apex Commerce API for cross-session persistence.
- On subsequent logins, auto-load saved profile and preferences.
- 'Reorder Last Meal' CTA displayed prominently on the home screen.

3.1.2 Action Items

#	Action Item	Details	Priority
A1.1	Build OTP Login Screen	Phone input + OTP verification UI. 30-second resend timer. WhatsApp API integration for OTP delivery.	P0
A1.2	Apex Commerce API Integration	POST /customers to create profile. GET /customers/{id} to fetch on login. Store: name, mobile, height, weight, goal, meal preferences.	P0
A1.3	Profile Creation Flow	Conditional — shown only on first login. Height/weight inputs → BMI auto-calculation. Goal picker (Weight Loss / Muscle Gain / Balanced Diet).	P0
A1.4	Session Management	JWT token stored in-device. Auto-expire after 30 min of inactivity. Auto-logout after order completion.	P0
A1.5	Reorder Last Meal	Fetch last order from Apex Commerce on login. Display 'Reorder' card on home. Add directly to cart with one tap.	P1

3.2 BMI-Based Meal Recommendation System

3.2.1 Functional Requirements

- Inputs: Height (cm), Weight (kg) → BMI = weight / (height in m)^2
- Goal selection: Weight Loss, Muscle Gain, Balanced Diet.
- System auto-generates a ranked list of 3–5 meal recommendations.
- Each meal card displays: Meal name, image, Calories, Protein (g), Carbs (g), Fat (g).
- Users can adjust portion size (S / M / L) within predefined limits. Macros update dynamically.
- Recommendation algorithm stored server-side (rules-based initially; ML in v2).

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

3.2.2 Action Items

#	Action Item	Details	Priority
A2.1	BMI Engine (Backend)	API endpoint: POST /recommendations. Input: {bmi, goal}. Output: ranked meal list with macros.	P0
A2.2	Recommendation Rules Config	Admin-configurable rules: BMI range → meal tags. E.g., BMI > 25 + Weight Loss → low-carb meals.	P0
A2.3	Meal Card UI Component	Reusable card: image, name, macro badges (cal/protein/carbs/fat), portion selector.	P0
A2.4	Portion Size Logic	Portion options per meal in admin. Dynamic macro recalculation on frontend.	P1
A2.5	Upsell Integration on Rec Screen	Show 1 upsell item (complementary meal/add-on) per recommendation. Track upsell conversion.	P1

3.3 Menu & Inventory Display

3.3.1 Functional Requirements

- Meals fetched in real-time from Rista POS inventory sync.
- Out-of-stock items automatically hidden or shown with an 'Unavailable' badge.
- Daily/Weekly Featured Meals section displayed at top of menu.
- Users can rate meals (1–5 stars) and save favorites to their profile.
- If a chosen item goes out of stock at checkout, suggest the most similar available meal.

3.3.2 Action Items

#	Action Item	Details	Priority
A3.1	Rista POS Sync Service	Webhook or polling (every 2 min) to sync inventory. Update meal availability in local DB.	P0
A3.2	Featured Meals Admin Toggle	Admin can pin up to 3 meals as 'Featured Daily' or 'Featured Weekly'.	P0
A3.3	Meal Rating & Favourites	POST /meals/{id}/rate. POST /customers/{id}/favourites. Displayed in profile and on meal cards.	P1
A3.4	Alternative Meal Suggestion Engine	When OOS at checkout: match by meal_tags, goal, calorie_range. Show 1–2 alternatives.	P0

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

3.4 Payments & Checkout

3.4.1 Functional Requirements

- UPI-only payments. No cards, no cash.
- Checkout displays a dynamic QR code generated via payment gateway (Razorpay/PhonePe).
- QR expires in 5 minutes. Auto-refresh on expiry.
- Discount code input field at checkout. Validated server-side.
- No referral codes. No printed receipts.
- Digital receipt sent via WhatsApp immediately on payment success.
- On payment failure: automated retry prompt → redirect to checkout screen. Max 3 retries.
- Corporate orders: bulk checkout from a single credential, centralized billing.

3.4.2 Action Items

#	Action Item	Details	Priority
A4.1	Payment Gateway Integration	Integrate Razorpay UPI or PhonePe Business. Webhook for payment status. Generate dynamic QR per order.	P0
A4.2	Checkout UI	Cart summary, discount field, QR display, countdown timer (5 min). Payment success / failure screens.	P0
A4.3	Discount Code Engine	POST /discounts/validate. Returns: {valid, discount_type, value}. Types: flat, percentage, free_item.	P1
A4.4	WhatsApp Receipt Service	On order.status = paid, trigger WhatsApp template message with order summary, amount, meal list.	P0
A4.5	Retry Logic	On payment failure: show retry CTA. After 3 failures, show 'Contact Staff' message. Log all failures.	P0
A4.6	Corporate Billing Module	Corporate accounts: single login, multi-meal cart, centralized invoice generation. Monthly PDF billing.	P2

3.5 Kiosk Hardware & Operational Reliability

3.5.1 Functional Requirements

- Self-restart mechanism on app crash or unresponsive state (watchdog process).
- Remote diagnostic integration (e.g., Splashtop, TeamViewer, or custom MDM solution).
- Touchscreen dead-zone detection with alert to admin panel.
- Printer errors surfaced to admin panel (note: no printer for receipts, but may be needed for kitchen labels).
- Auto-lock screen and return to idle after 60 seconds of inactivity.

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

3.5.2 Action Items

#	Action Item	Details	Priority
A5.1	Watchdog / Self-Restart	Background daemon monitors app heartbeat. Restart app if no response in 30 sec. iOS: use Guided Access + MDM policy.	P0
A5.2	MDM Integration	Enroll kiosk in MDM (e.g., Jamf, Mosyle for iPad). Remote lock, update, diagnostic access.	P0
A5.3	Idle Screen / Attract Loop	After 60 sec inactivity: show branded idle screen with 'Tap to Order' CTA.	P1
A5.4	Hardware Status Monitoring	POST hardware alerts to admin dashboard. Alert types: screen_unresponsive, app_crashed, network_down.	P1
A5.5	Network Offline Handling	Show 'Network Unavailable' screen. Block ordering. Auto-resume when connection restored.	P0

4. Module B — Website & Meal Subscription Storefront

Framework: Next.js | Deployment: Vercel / AWS | CMS: Headless or custom Admin

4.1 Public Website Pages

4.1.1 Functional Requirements

- Branding fully customizable (colors, fonts, logo) via theme config.
- Landing Page: Hero section, benefits, how-it-works, meal plan highlights, CTA.
- Meal Plans Listing Page: Card grid of available plans with key info.
- Individual Meal Plan Detail Page (optional per plan): Full description, ingredients, nutritional info, pricing.
- Contact Us Page: Form with name, email, phone, message. Backend stores submissions.
- About Us Page: Static content managed via admin.
- Legal Pages: Terms & Conditions, Privacy Policy (rich text, admin-editable).

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

4.1.2 Action Items

#	Action Item	Details	Priority
B1.1	Next.js Project Setup	Configure SSR, routing, theme provider. Integrate Tailwind CSS.	P0
B1.2	Landing Page Development	Hero, benefits, how-it-works, plan preview cards, footer. Mobile responsive.	P0
B1.3	Meal Plans Listing & Detail Pages	Dynamic routes /plans and /plans/[slug]. Fetch from meals admin API.	P0
B1.4	Contact & About Pages	Contact form with server-side validation and email/DB storage. Static About page.	P1
B1.5	Legal Pages CMS	Admin-editable rich text for T&C, Privacy Policy. Versioned updates.	P1
B1.6	SEO & Meta Tags	Dynamic OG tags per page. Sitemap.xml, robots.txt generation.	P1

4.2 Subscription Purchase Flow

4.2.1 Functional Requirements

- Customer selects a meal plan → chooses options (meal preferences, add-ons, delivery slot).
- Different plans can have different options/add-ons as configured in admin.
- Payment via UPI / Cards / Netbanking (website is not kiosk-constrained).
- On successful payment, subscription is activated in customer account.
- Subscription auto-renews unless cancelled. Renewal reminder via WhatsApp/email.

4.2.2 Action Items

#	Action Item	Details	Priority
B2.1	Plan Selection & Configurator UI	Multi-step flow: Plan → Preferences → Add-ons → Delivery Slot → Payment.	P0
B2.2	Dynamic Options Engine	Admin configures options per plan. Frontend renders options dynamically from API.	P0
B2.3	Payment Integration (Web)	Razorpay / Stripe: UPI, cards, netbanking. Webhook for subscription activation.	P0
B2.4	Subscription Lifecycle Management	DB model: subscriptions table with status, renewal_date, plan_id. Cron for renewals.	P0
B2.5	Renewal Reminders	WhatsApp & email reminder 3 days before renewal. Allow 1-click renewal or cancel.	P1

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

4.3 Customer Dashboard (Web)

4.3.1 Functional Requirements

- Authenticated area (login with mobile OTP or email/password).
- View and manage active meal subscriptions (pause, cancel, upgrade).
- Wallet & Ledger: balance, top-up history, deduction history.
- View and edit customer profile (name, address, dietary preferences).

4.3.2 Action Items

#	Action Item	Details	Priority
B3.1	Customer Auth (Web)	Mobile OTP + optional email/password. JWT session. Persistent login.	P0
B3.2	Subscription Management UI	List active subs. Actions: Pause (with date range), Cancel (with reason), Upgrade/Downgrade.	P0
B3.3	Wallet & Ledger	Display current balance. Transaction history table (paginated). Top-up flow via payment gateway.	P1
B3.4	Profile Edit	Editable: name, phone, email, delivery addresses (multiple), dietary preferences.	P0

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

5.2 Operations Admin

#	Action Item	Details	Priority
C2.1	Daily Meal Processing Flow	Ops dashboard: today's orders, subscription meals to prepare. Status workflow: Pending → Preparing → Ready → Delivered.	P0
C2.2	Kitchen Print Documents	Generate print-ready PDFs: Kitchen Prep Sheet (by dish), Packing Label (customer name, meal, slot), Dispatch Manifest.	P0
C2.3	Detailed Order Info	Drill into any order: customer profile, items, macros, special instructions, payment status.	P0
C2.4	Rista POS Sync Dashboard	Show last sync time, sync status, inventory levels. Manual sync trigger button.	P1

5.3 Business Configuration & Staff Management

#	Action Item	Details	Priority
C3.1	Kitchen & Hub Setup	CRUD for kitchens and delivery hubs. Map delivery radius per hub (polygon or radius).	P1
C3.2	Hub-Based Pricing	Configure delivery charges per hub. Override meal prices per hub if needed.	P1
C3.3	Customer Management	Search/view customers. Subscription history, wallet balance, contact info. Manual wallet top-up.	P0
C3.4	Staff Management	CRUD for staff accounts. Roles: Super Admin, Ops Manager, Kitchen Staff, Delivery Agent.	P1
C3.5	Analytics Dashboard	Daily sales (revenue, orders, AOV). Meal popularity. Customer return rate. Upsell conversion. Export to CSV.	P1
C3.6	Corporate Account Management	Create corporate accounts. Assign bulk meal plans. Configure centralized billing. Generate monthly invoices.	P2

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

6. Module D — Mobile Applications (iOS & Android)

Framework: React Native | Shared codebase | Deployment: App Store + Google Play

6.1 Core Mobile Features

#	Action Item	Details	Priority
D1.1	App Setup & Auth	React Native project. Mobile OTP login. Push notification setup (FCM/APNs).	P0
D1.2	Browse & Subscribe to Meal Plans	Replicate web subscription flow. Mobile-optimized multi-step configurator.	P0
D1.3	Skip a Meal	From active subscription: select dates to skip. Advance notice cutoff enforced (e.g., before 8 AM for same day).	P0
D1.4	Switch Delivery Location	Toggle between 'Home' and 'Work' addresses per delivery. Per-order or set as default.	P0
D1.5	Add-On Selection	Before daily meal delivery, customers can add drinks/snacks. Cutoff time enforced.	P1
D1.6	Subscription Dashboard (Mobile)	View active plan, upcoming meals, next delivery date, wallet balance.	P0
D1.7	Push Notifications	Order confirmed, meal dispatched, delivery ETA, renewal reminder, skip confirmation.	P1
D1.8	App Store Submission Prep	App icons, screenshots, store descriptions, privacy policy URL, review guidelines compliance.	P1

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

7. Core Data Models

7.1 Key Entities

Entity	Key Fields
Customer	id, name, mobile, email, height, weight, bmi, goal, preferences[], wallet_balance, apex_commerce_id
Meal	id, name, description, image_url, tags[], calories, protein, carbs, fat, base_price, portions{S,M,L}, is_available, is_featured
Subscription Plan	id, name, duration_days, price, meals[], delivery_slots[], options_config, add_ons[]
Subscription	id, customer_id, plan_id, status, start_date, end_date, renewal_date, delivery_address_id
Order	id, customer_id, type(kiosk|web|mobile), items[], total, discount_code, payment_status, payment_ref, created_at
Delivery	id, subscription_id, date, meal_id, add_ons[], location_type(home|work), status, kitchen_id
Inventory	meal_id, quantity, last_sync_at, rista_pos_ref
Staff	id, name, email, role(super_admin|ops|kitchen|delivery), kitchen_id
Corporate Account	id, company_name, billing_email, credit_limit, meal_plan_ids[], contact_person

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

8. API Endpoints Summary

8.1 Authentication

Method	Endpoint	Description
POST	/auth/otp/send	Send OTP to mobile via WhatsApp/SMS
POST	/auth/otp/verify	Verify OTP, return JWT
POST	/auth/logout	Invalidate session

8.2 Customers & Profiles

Method	Endpoint	Description
POST	/customers	Create customer profile (Apex Commerce sync)
GET	/customers/:id	Fetch customer profile
PATCH	/customers/:id	Update profile / preferences
GET	/customers/:id/orders/last	Fetch last order for Reorder feature
GET	/customers/:id/favourites	Fetch saved favourite meals
POST	/customers/:id/favourites	Add meal to favourites

8.3 Meals & Recommendations

Method	Endpoint	Description
GET	/meals	List all available meals (with inventory status)
GET	/meals/:id	Get meal detail
POST	/recommendations	Get BMI + goal-based meal recommendations
GET	/meals/featured	Get featured daily/weekly meals
POST	/meals/:id/rate	Submit meal rating

8.4 Orders & Payments

Method	Endpoint	Description
POST	/orders	Create order
GET	/orders/:id	Get order status
POST	/orders/:id/payment/initiate	Generate UPI QR for order
POST	/payments/webhook	Handle payment gateway webhook
POST	/discounts/validate	Validate discount code

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

9. Phased Development Roadmap

Phase	Duration	Deliverables
Phase 1 — Foundation	Weeks 1–3	Backend API setup, DB schema, Rista POS sync, OTP auth, Apex Commerce integration
Phase 2 — Kiosk MVP	Weeks 4–7	Kiosk app: login, BMI recommendation, menu browsing, UPI checkout, WhatsApp receipt, hardware reliability
Phase 3 — Web MVP	Weeks 6–9	Website landing pages, subscription purchase flow, customer dashboard (web)
Phase 4 — Admin Panel	Weeks 8–11	Meals CRUD, ops flow, kitchen printing, analytics dashboard, staff management
Phase 5 — Mobile Apps	Weeks 10–14	React Native apps: subscribe, skip, switch location, add-ons, push notifications
Phase 6 — Advanced Features	Weeks 14–18	Corporate billing, loyalty program, ML-based recommendations, QR pickup system
Phase 7 — QA & Launch	Weeks 17–20	Full integration testing, UAT, performance testing, App Store submission, go-live

10. QR-Based Pickup System

For pre-ordered meals (corporate and subscription), a QR-based collection system will be implemented:

- On order confirmation, generate a unique QR code and send via WhatsApp.
- At the pickup point, a staff-facing app (or kiosk camera) scans the QR.
- System validates QR → marks order as Collected → notifies customer.
- QR is single-use and expires 4 hours after the meal's ready time.

#	Action Item	Details	Priority
E1.1	QR Generation Service	Generate unique, time-limited QR per order on payment success. Store QR token in DB.	P1
E1.2	QR Delivery via WhatsApp	Send QR image via WhatsApp Business API on order ready.	P1
E1.3	Staff Scanner Interface	Simple web/app screen for staff to scan QR with device camera. Show order summary on scan.	P1
E1.4	QR Validation API	POST /orders/qr/scan. Validate token, check expiry, mark as collected.	P1

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.

11. Non-Functional Requirements

Requirement	Specification
Performance	API response time < 200ms (p95). Kiosk screen transitions < 500ms.
Availability	Backend uptime > 99.5%. Kiosk app offline-safe for browsing (online required for payment).
Security	HTTPS everywhere. JWT with expiry. OTP rate limiting (max 5 attempts / 10 min). PII encrypted at rest.
Scalability	Stateless API designed for horizontal scaling. Redis for session/cache.
Data Retention	Orders retained for 3 years. Customer profiles until account deletion request.
Compliance	GDPR-style data handling. Customer data deletion on request within 30 days.
Accessibility	Kiosk UI: minimum 44px touch targets. High contrast mode for outdoor lighting.
Localization	English only (Phase 1). Architecture must support i18n for future language additions.

12. Open Questions & Decisions Required

#	Question	Owner	Status
Q1	Confirm payment gateway: Razorpay vs PhonePe Business for kiosk UPI.	Business	Open
Q2	Confirm MDM solution for iPad kiosk management (Jamf vs Mosyle vs custom).	Tech Lead	Open
Q3	Is printer needed for kitchen labels at kiosk? (Affects hardware spec)	Operations	Open
Q4	Loyalty points program: structure (points per rupee? flat per order?) and redemption rules.	Business	Open
Q5	Corporate billing cycle: monthly invoice via email or portal download?	Business	Open
Q6	Subscription skip advance cutoff time (e.g., must skip before 8 AM for same day).	Operations	Open
Q7	WhatsApp Business Account approval status and WABA ID/Phone Number confirmation.	Business	Open
Q8	Define 'similar meal' logic for OOS alternatives — by tags, macros, or price?	Product	Open

Confidential: This document is the property of Reachvel and is intended solely for the designated recipient.
Unauthorized use, sharing, or reproduction is strictly prohibited.